Lab 6.2: Prime Check (F# Functional Style)

lab6.2.fsx

```
let isPrime (n: int) : bool =
    let rec hasNoDivisor d =
        if d * d > n then true
        elif n % d = 0 then false
        else hasNoDivisor (d + 1)
    n >= 2 && hasNoDivisor 2

let describe (n: int) : string =
    sprintf "%d is %s" n (if isPrime n then "prime" else "not prime")

let testSets =
    [ "Primes", [ 2; 3; 17; 97 ]
      "Composites", [ 4; 15; 49; 100 ]
      "Edge cases", [ -7; 0; 1 ] ]

testSets
|> List.iter (fun (name, numbers) ->
    printfn "%s:" name
    numbers |> List.map describe |> List.iter (printfn "  %s"))

let primesUpTo30 = [ 1 .. 30 ] |> List.filter isPrime
printfn "Primes from 1 to 30: %A" primesUpTo30
```